

EDWARD BAKER

MACHINE LEARNING ENGINEER | SOFTWARE ENGINEER | DATA SCIENTIST

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Production-focused engineer building machine learning models, cloud pipelines, APIs, and reusable Python systems.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++, JavaScript, R

ML & Data: Scikit-learn, Azure Machine Learning, Snowflake, Streamlit

Cloud & Engineering: Azure, AWS, Docker, Kubernetes/AKS, Azure DevOps, Git, Linux

PROFESSIONAL EXPERIENCE

ELEVATE CREDIT | Data Scientist II (promoted Nov 2025) | Hybrid

June 2024 - Present

- Engineered an end-to-end Azure ML pipeline to reparameterize eight production credit-risk models monthly, reducing model refresh time from two weeks to one day and enabling quantitative drift monitoring.
- Developed and productionized a risk of chargeoff (ROC) collections model, beating existing third-party vendor solution performance and allowing the company to move our collections models in-house saving \$100K+ annually.
- Developed and productionized an XGBoost credit-risk model that improved AUC by 13%; built a modular ~1,200-line Snowflake SQL layer to transform raw XML into validated training datasets.
- Designed and deployed Dockerized model inference APIs on Azure Kubernetes Service, incorporating unit tests and standardized deployment workflows for production reliability.
- Built and maintained an internal Python package with 50+ reusable modules for feature engineering, validation, and lifecycle management; adopted by a 15-member team across 12+ production projects.
- Implemented a Snowflake feature store containing 50+ proprietary engineered attributes, replacing ad hoc notebook logic with reusable, governed features.
- Led adoption of Azure ML engineering practices, including versioned data assets, registered environments and models, and reproducible training and deployment pipelines.

XYLEM INC. | Software Engineer Intern | Raleigh, NC

May 2022 - Aug 2022

- Developed three Angular application screens for an enterprise data-lake product and integrated customer data using SQL and AWS.
- Built Python scripts supporting enterprise security analysis for 1,000+ employees.

UNC CHAPEL HILL - COMPUTER SCIENCE | Undergraduate Learning Assistant | Chapel Hill, NC

Aug 2022 - May 2023

- Designed and taught an MVC architecture module in COMP 301 for 50+ students and provided Java debugging support focused on object-oriented design and software engineering principles. Assisted with exam question development and grading.

SELECTED TECHNICAL PROJECT

ARDUINO MIDI LIGHT SHOW | Personal Project

July 2025

- Engineered a real-time, event-driven system using Python, multithreading, serial I/O, and embedded C++ to synchronize LED hardware with audio playback.
- Implemented a custom low-latency serial protocol and extensible architecture for dynamically mapping MIDI attributes to lighting patterns.

EDUCATION, LEADERSHIP & CERTIFICATION

M.S. Analytics, NC State University | May 2024

B.S. Computer Science, Minor in Religious Studies, UNC Chapel Hill | May 2023

Leadership & Honors: Elected student team lead for U.S. Department of Education practicum; later selected as Elevate technical liaison for the 2026 practicum cohort; graduated with distinction from UNC Chapel Hill, member of the Honors college; HackNC 2022 Best Junior Hackers winner.

Certification: AWS Certified Cloud Practitioner